

RoboPet - Phase 4 Build Guide

(Placeholder ile Oyun)

Bu rehber hic sprite uretmeden tum Living Room dekor + magaza dongusunu placeholder kutularla kurur. Gercek PNG dosyalari sonra assets/images/decor/<id>.png icine atilince otomatik gercek sprite olur - kod degismez.

Build sirasi

- 1. Veri katmani - DecorDef + enum + katalog
- 2. Kayit katmani - Isar PlacedDecor + DecorRepository (build_runner)
- 3. Gorsel - SimpleFloor + DecorComponent (placeholder)
- 4. Dunya - LivingRoomWorld (duvar + zemin + kopek + yerlesik dekor)
- 5. UI - Decorate modu + Magaza overlay
- 6. Test checklist

1) Veri katmani - DecorDef + katalog

```
enum DecorCategory { seating, screen, plant, lighting, storage, rug, wallArt }
```

```
class DecorDef {
  const DecorDef({
    required this.id,
    required this.name,
    required this.asset,
    required this.category,
    required this.currency,
    required this.price,
    required this.defaultSize,
    required this.defaultPos,
    this.layer = 0,
  });
  final String id;
  final String name;
  final String asset;          // decor/<id>.png  (yoksa placeholder)
  final DecorCategory category;
  final CurrencyType currency;
  final int price;
  final Vector2 defaultSize;
  final Vector2 defaultPos;
  final int layer;
}

final List<DecorDef> kLivingRoomDecor = [
  DecorDef(id: 'couch', name: 'Neon Couch', asset: 'decor/couch.png',
    category: DecorCategory.seating, currency: CurrencyType.scrap, price: 120,
    defaultSize: Vector2(150, 70), defaultPos: Vector2(0, 300), layer: 0),
  DecorDef(id: 'holo_tv', name: 'Holo TV', asset: 'decor/holo_tv.png',
    category: DecorCategory.screen, currency: CurrencyType.premiumGems, price: 15,
    defaultSize: Vector2(140, 90), defaultPos: Vector2(0, -120), layer: -2),
  DecorDef(id: 'plant', name: 'Potted Plant', asset: 'decor/plant.png',
```

```

        category: DecorCategory.plant, currency: CurrencyType.scrap, price: 60,
        defaultSize: Vector2(50, 80), defaultPos: Vector2(140, 320), layer: 1),
    DecorDef(id: 'lamp', name: 'Floor Lamp', asset: 'decor/lamp.png',
        category: DecorCategory.lighting, currency: CurrencyType.scrap, price: 80,
        defaultSize: Vector2(40, 110), defaultPos: Vector2(-150, 320), layer: 1),
    DecorDef(id: 'rug', name: 'Round Rug', asset: 'decor/rug.png',
        category: DecorCategory.rug, currency: CurrencyType.scrap, price: 40,
        defaultSize: Vector2(220, 70), defaultPos: Vector2(0, 340), layer: -5),
    DecorDef(id: 'neon_sign', name: 'Neon Sign', asset: 'decor/neon_sign.png',
        category: DecorCategory.wallArt, currency: CurrencyType.premiumGems, price: 10,
        defaultSize: Vector2(90, 40), defaultPos: Vector2(120, -180), layer: -3),
];

DecorDef decorById(String id) => kLivingRoomDecor.firstWhere((d) => d.id == id);

```

2) Kayit katmani - Isar PlacedDecor + Repository

```

import 'package:isar/isar.dart';
part 'placed_decor.g.dart';

@collection
class PlacedDecor {
  Id isarId = Isar.autoIncrement;
  @Index() late String itemId;
  late String room;
  late double x;
  late double y;
  double scale = 1.0;
  bool flipped = false;
  int layer = 0;
}

class DecorRepository {
  DecorRepository(this._isar);
  final Isar _isar;

  Future<List<PlacedDecor>> forRoom(String room) =>
    _isar.placedDecors.filter().roomEqualTo(room).findAll();

  Future<int> place(PlacedDecor d) =>
    _isar.writeTxn(() => _isar.placedDecors.put(d));

  Future<void> update(PlacedDecor d) =>
    _isar.writeTxn(() => _isar.placedDecors.put(d));

  Future<void> remove(int isarId) =>
    _isar.writeTxn(() => _isar.placedDecors.delete(isarId));
}

final decorRepositoryProvider = Provider<DecorRepository>(
  (ref) => DecorRepository(ref.watch(isarProvider)),
);

```

NOT: PlacedDecorSchema yi Isar.open([...]) semalarina ekle, sonra: dart run build_runner build -d

3) Gorsel - SimpleFloor + DecorComponent

```

class SimpleFloor extends PositionComponent {
  SimpleFloor({required Vector2 size, required Vector2 position})

```

```

        : super(size: size, position: position, priority: -9);
final _floor = Paint()..color = const Color(0xFF161A2E);
final _shade = Paint()..color = const Color(0x22000000);
final _edge = Paint()..color = const Color(0xAA22E1FF)..strokeWidth = 2;
@override
void render(Canvas canvas) {
    final w = size.x, h = size.y;
    canvas.drawRect(Rect.fromLTWH(0, 0, w, h), _floor);
    canvas.drawRect(Rect.fromLTWH(0, h * 0.5, w, h * 0.5), _shade);
    canvas.drawLine(const Offset(0, 0), Offset(w, 0), _edge);
}
}

class DecorComponent extends SpriteComponent
    with DragCallbacks, HasGameReference<RoboPetGame> {
    DecorComponent({required this.def, required this.placed})
        : super(anchor: Anchor.bottomCenter, priority: def.layer);
    final DecorDef def;
    final PlacedDecor placed;
    bool editable = false;

    @override
    Future<void> onLoad() async {
        size = def.defaultSize * placed.scale;
        position = Vector2(placed.x, placed.y);
        if (placed.flipped) flipHorizontally();
        try {
            sprite = await Sprite.load(def.asset); // gercek PNG varsa kullan
        } catch (_) {
            // PLACEHOLDER: renkli kutu + isim
            add(RectangleComponent(
                size: size,
                paint: Paint()..color = const Color(0xFF394067),
            ));
            add(TextComponent(
                text: def.name,
                anchor: Anchor.center,
                position: size / 2,
                textRenderer: TextPaint(
                    style: const TextStyle(color: Color(0xFF9FB4FF), fontSize: 11),
                ),
            ));
        }
    }

    @override
    bool containsLocalPoint(Vector2 p) => editable && super.containsLocalPoint(p);

    @override
    void onDragUpdate(DragUpdateEvent event) {
        if (!editable) return;
        position += event.localDelta;
        position.x = position.x.clamp(-180, 180);
        position.y = position.y.clamp(-300, 345);
    }

    @override
    void onDragEnd(DragEndEvent event) {
        if (!editable) return;
        placed
            ..x = position.x
            ..y = position.y;
    }
}

```

```

        game.onDecorMoved(placed);    // kalici yap
    }
}

```

4) Dünya - LivingRoomWorld

```

class LivingRoomWorld extends World with HasGameReference<RoboPetGame> {
    late final RobotDog dog;

    @override
    Future<void> onLoad() async {
        // duvar (sprite yoksa Cursor gecici renk koyabilir)
        add(SpriteComponent(
            sprite: await Sprite.load('rooms/living_wall.png'),
            size: Vector2(400, 350),
            position: Vector2(-200, -350),
            priority: -10,
        ));
        // zemin (tamamen kod)
        add(SimpleFloor(size: Vector2(400, 350), position: Vector2(-200, 0)));

        // kayitli dekorlar
        final repo = game.ref.read(decorRepositoryProvider);
        for (final p in await repo.forRoom('living')) {
            add(DecorComponent(def: decorById(p.itemId), placed: p));
        }

        // kopek
        dog = RobotDog(position: Vector2(0, 280));
        add(dog);
    }

    Future<void> addDecor(DecorDef def) async {
        final repo = game.ref.read(decorRepositoryProvider);
        final placed = PlacedDecor()
            ..itemId = def.id
            ..room = 'living'
            ..x = def.defaultPos.x
            ..y = def.defaultPos.y
            ..layer = def.layer;
        placed.isarId = await repo.place(placed);
        add(DecorComponent(def: def, placed: placed));
    }

    void setEditMode(bool on) {
        children.whereType<DecorComponent>().forEach((d) => d.editable = on);
        if (on) { dog.stopWander(); } else { dog.startWander(); }
    }
}

```

5) UI - Decorate modu + Magaza overlay

```

// RoboPetGame icinde
bool decorateMode = false;

void toggleDecorate() {
    decorateMode = !decorateMode;
    (world as LivingRoomWorld).setEditMode(decorateMode);
    if (decorateMode) { overlays.add('shop'); } else { overlays.remove('shop'); }
}

```

```

Future<void> onDecorMoved(PlacedDecor p) =>
    ref.read(decorRepositoryProvider).update(p);

class ShopOverlay extends ConsumerWidget {
    const ShopOverlay({super.key, required this.game});
    final RoboPetGame game;

    @override
    Widget build(BuildContext context, WidgetRef ref) {
        final economy = ref.watch(economyProvider);
        return Align(
            alignment: Alignment.bottomCenter,
            child: Container(
                height: 180,
                color: const Color(0xCC0B0E1A),
                child: ListView(
                    scrollDirection: Axis.horizontal,
                    padding: const EdgeInsets.all(12),
                    children: kLivingRoomDecor.map((d) {
                        final owned = economy.ownsDecor(d.id);
                        return _ShopCard(
                            def: d,
                            owned: owned,
                            onTap: () async {
                                final world = game.world as LivingRoomWorld;
                                if (owned) {
                                    await world.addDecor(d);
                                } else {
                                    final ok = await ref.read(economyProvider.notifier)
                                        .buyDecor(d.id, d.currency, d.price);
                                    if (ok) await world.addDecor(d);
                                }
                            },
                        );
                    }).toList(),
                ),
            ),
        );
    }
}

// economy notifier
Future<bool> buyDecor(String id, CurrencyType c, int amount) async {
    final ok = await spend(c, amount, reason: 'decor:' + id); // economyVersion++
    if (ok) await grantDecor(id);
    return ok;
}

```

NOT: overlays haritasina shop: (ctx, game) => ShopOverlay(game: game) ekle. _ShopCard basit bir Container + isim + fiyat + (sahipse Yerlestir) etiketi.

6) Test checklist

- Decorate butonu -> magaza aciliyor, kopek duruyor
- Scrap ile couch al -> ekrana isimli placeholder kutu dusuyor
- Kutuyu surukle -> sinirlar icinde kaliyor (x +-180, y -300..345)
- Decorate kapat -> tekrar ac -> kutu yerinde

- Uygulamayı kapat-ac (restart) -> dekor kalici (Isar)
- Yetersiz para -> satın alma reddediliyor
- decor/couch.png ekle -> restart -> placeholder yerine gercek sprite (kod degismeden)